Mohammad Seraj

+91-9628104350 mail.serajansari@gmail.com LinkedIn GitHub Portfolio Varanasi, UP, India

Professional Summary

AI/ML Engineer with 1.5 years of experience designing and deploying NLP, Generative AI, and deep learning solutions. Built production RAG systems, LLM fine-tuning pipelines, and scalable AI services using LangChain, PyTorch, FastAPI, and AWS across healthcare and enterprise domains.

Work Experience

Freelance AI Engineer / Independent Research Nov 2025 – Present
Self-Employed *Remote*

- Co-authored manuscript (ct-2026-00508r) -*Quantum-Physics Inspired ML Model for Partition Coefficient Prediction* submitted to **Journal of Chemical Theory and Computation** (under review).

AI Developer Apr 2025 – Oct 2025
Anervea Data Labs Private Limited *Pune, Maharashtra, India*

- **Project: Clinical Trial Patient Recruitment Platform (GenAI / RAG System)**
- Architected and deployed an AI-powered system that reduced patient identification time from **several months to under 30 minutes**, automating end-to-end eligibility screening from EHR and claims data at scale.
- Engineered end-to-end **NLP ingestion pipelines** using AWS Textract and AWS Bedrock (Claude Sonnet 3.5) to extract and structure clinical trial protocol components trial titles, inclusion/exclusion criteria into standardized, machine-readable formats, enabling automated protocol parsing across **multiple concurrent trials**.
- Designed and implemented a **RAG architecture with PGVector**, enabling semantic retrieval of ICD-10, CPT, and LOINC medical codes alongside demographic and gender-specific eligibility criteria for precise patient matching, reducing manual review effort by **˜70%**.
- Integrated with **Komodo Workspace** (HIPAA-compliant healthcare dataset) to operationalize cohort generation surfacing each matched patient's most recent HCOs and HCPs, enabling researchers to directly contact and recruit eligible volunteers without exposing protected patient data.
- Implemented **LLM evaluation and monitoring pipelines** for RAG workflows using **RAGAS metrics** (faithfulness, answer relevance, context precision) and **LangSmith tracing** to analyze retrieval quality, prompt behavior, and end-to-end response reliability.
- Deployed **HIPAA-compliant microservices** on AWS (EC2, S3) and Snowflake, enabling near real-time trial feasibility analysis and accelerating time-to-market for novel therapies.
- **Project: BioBERT Fine-Tuning for Clinical NLP Classification**
- Fine-tuned **BioBERT** for clinical-trial NLP classification using PEFT/QLoRA with 4-bit quantization via bitsandbytes, achieving efficient low-memory training and improved domain-adapted inference performance.
- Containerized all NLP and GenAI services with **Docker**; managed CI/CD workflows via Jenkins and GitHub, ensuring reproducible and scalable cloud deployments.
- *Stack: Python, LangChain, LangGraph, FastAPI, AWS (EC2, S3, Textract, Bedrock), Vector Databases (PGVector), RAG Pipelines, LLM Evaluation (RAGAS), LangSmith Monitoring, Snowflake, Docker, Jenkins, BioBERT, Hugging Face Transformers, PEFT/QLoRA, bitsandbytes, PyTorch, CUDA*

Research Assistant Jun 2024 – Apr 2025
NeuroCare-eLab *New Delhi, India*

- Led computational drug discovery research on **LogP prediction and molecular solubility estimation**, combining classical and quantum descriptors (QM, QMPSA, ΔG) to engineer highly predictive molecular feature sets.
- Benchmarked GNN, XGBoost, Random Forest, and ExtraTrees models with 5-fold cross-validation; achieved state-of-the-art LogP performance with **GNN (R^2=0.959, MAE=0.115, MSE=0.222)**, outperforming XGBoost (R^2=0.932) and feature-augmented ExtraTrees (R^2=0.849).
- Designed end-to-end computational pipelines covering data preprocessing, model training, evaluation, and documentation, enabling reproducible experimentation that accelerated drug discovery workflows.
- *Stack: Python, PyTorch, scikit-learn, GNNs, RDKit, Modred, DeepChem, Feature Engineering (Classical + Quantum Descriptors)*

Researcher — M.Tech Thesis Jun 2023 – May 2024
Indraprastha Institute of Information Technology Delhi (IIIT-D) *New Delhi, India*

- Engineered a **volumetric 3D deep learning model** on `.nii` MRI files, improving Alzheimer's diagnostic accuracy by **64% over 2D baselines**; Thesis Link.
- Developed multi-view YOLOv8n classifiers across axial, coronal, and sagittal planes surpassing VGG16 baselines (coronal 80.5%, sagittal 78.1%, axial 71.0%), and built an ensemble achieving **75% overall accuracy**.
- *Stack: PyTorch, TensorFlow, YOLOv8n, OpenCV*

Projects

Autonomous Voice-Enabled Dealership Agent View Code

CrewAI, LangChain, Llama 3.3, FastAPI, Deepgram, Docker

- Built a multi-agent orchestration system (CrewAI + Llama 3.3) automating customer-service workflows: test-drive scheduling, intent detection, and knowledge-base retrieval — **replacing manual handling of repetitive customer queries end-to-end**.
- Engineered a low-latency voice pipeline with Deepgram Nova-2 (STT) and Aura (TTS); implemented tool-level guardrails and strict prompt controls that **eliminated date/time booking hallucinations** to achieve near-zero error rate in scheduling logic.
- Implemented an async FastAPI backend managing concurrent audio streams, session chat history, and dynamic knowledge retrieval at production scale, supporting **multiple simultaneous voice sessions** without degradation.

GenAI Document Intelligence Platform View Code

Python, LangChain, Hugging Face, FastAPI, Qdrant, Docker

- Designed an end-to-end RAG pipeline for intelligent document understanding, semantic search, and QA over enterprise document corpora.
- Containerized inference services via FastAPI and Docker; iterative prompt optimization improved answer accuracy and reduced hallucination rate in production.

Education

Indraprastha Institute of Information Technology, Delhi (IIIT-D) 2022 – 2024

M.Tech in Computational Biology — CGPA: 7.96 / 10.0 *New Delhi, India*

- *Relevant Coursework: Natural Language Processing, Machine Learning in Biomedical Applications, Big Data Mining in Healthcare, Biomedical Image Processing*

Hindustan College of Science and Technology, Mathura 2018 – 2022

B.Tech in Biotechnology — CGPA: 7.80 / 10.0 *Mathura, UP, India*

Technical Skills

Core Strengths: RAG Systems, Large Language Models, Agentic AI, LLM Fine-Tuning, NLP, Production ML Pipelines

Generative AI / LLMs: LangChain, LangGraph, CrewAI, Prompt Engineering, Tool Calling, PEFT (QLoRA), 4-bit Quantization, Embeddings, AWS Bedrock

NLP: BioBERT, Hugging Face Transformers, NER, Text Classification, Tokenization, spaCy, NLTK, Document Understanding

ML / Deep Learning: Neural Networks, GNN, XGBoost, Random Forest, SVM, Cross-Validation, Feature Engineering, Hyperparameter Tuning

Cloud & MLOps: AWS (EC2, S3, Textract, Bedrock), Docker, FastAPI, CI/CD, Jenkins, GitHub Actions, MLflow, Model Serving

Databases & Vector Stores: PGVector, Qdrant, FAISS, Snowflake, SQL

Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Streamlit, OpenCV, RDKit, DeepChem

Languages & Tools: Python, SQL, Git, GitHub, VS Code, Google Colab, CUDA, cuDNN

Achievements

DBT GAT-B (2022) — All India Rank 133

Department of Biotechnology, Government of India

- Cleared the national biotechnology aptitude examination conducted by the Department of Biotechnology, Government of India.